

Hypothesis Testing: One Population

MGMT 670



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Part I: Key Concepts (Minimal Math)

- Background of hypothesis testing
- Null and alternative hypothesis
- P-value to measure the significance of evidence
- Overview of hypothesis testing steps
- Errors and power of hypothesis

[Data Science Interview Questions](#)

[Data Science Interview Questions 2](#)

[Data Science Interview Questions 3](#)

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Example 1: Processing a Loan Application

- You're the commercial loan officer at a bank, in the process of reviewing a loan application recently filed by a local firm.
 - The largest single asset of the firm is \$3 million in accounts receivable.
 - You send a junior associate, Mary, out to meet with the firm's credit manager to check whether these receivables actually exist.

Mary's report

- The firm's manager told her
 - There were 10,000 customers holding credit accounts with the firm.
 - The average amount due to be paid per account was at least \$300.
- She confirmed (by physical measurement) the existence of about 10,000 customer folders. (You decide to accept this part of the credit manager's claim.)
- How to check the claim: The average amount due to be paid per account was at least \$300?

- Mary selected a random sample of 8 accounts, and contacted the account-holders. They all acknowledged soon-to-be-paid debts to the firm. Mary looks at the sample mean.
 - If it is **well below** \$300 (maybe $< \$200$), then this claim is suspicious to her
 - If it is **well above** \$300 (maybe $> \$400$), then she believes the claim
 - Mary doesn't know what to do if the sample mean is close to \$300. She reports back to you, let you decide!
- We are now helping Mary to make **more accurate decision**.

The Goal of Hypothesis Testing

To put a single phrase which summarizes all that the data says with respect to the original statement.

The data, all by itself, makes me _____ suspicious, because the data, all by itself, _____ contradicts the statement.”
{not at all, a little bit, quite, very much}

We wish to choose the phrase which best fills the blanks.

Hypothesis Testing

A statement has been made. We must decide whether to believe it (or not). Our belief decision must ultimately stand on three legs:

- What does our general background knowledge and experience tell us (for example, what is the reputation of the speaker)?
- What is the cost of being wrong (believing a false statement, or disbelieving a true statement)?
- **What does the relevant data tell us?**

To do that: compute p-value *(the significance level of the data)*

P-value is the probability of observing data as extreme as, or more extreme than, the data actually observed, under the assumption that the statement is true.

If P-value is small, what's our conclusion?

Interpret the p-value: A small value forces us to say, “Either **the statement is true and we've been very unlucky** (i.e., we've drawn a very misrepresentative sample), or **the statement is false**. We don't typically expect to be very unlucky, so the data, all by itself, makes us suspicious.”

This is Similar to the British System of Criminal Justice

- The statement on trial – the so-called “null hypothesis” – is that “the accused is innocent.”
- The prosecution presents evidence.
- The jury asks itself: “How likely is it that this evidence would have turned up, just by chance, if the accused really *is* innocent?”
- If this probability is close to 0, then the evidence strongly contradicts the initial presumption of innocence ... and the jury finds the accused “Guilty!”

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Null and Alternative Hypotheses

- Null Hypothesis, H_0 (**being innocent**)
 - The average amount due to be paid per account was at least \$300.
 - An assertion about a population parameter
 - True until we have sufficient evidence to conclude otherwise
- Alternative Hypothesis, H_a (**being guilty**)
 - The average amount due to be paid per account was less than \$300.
 - Opposite of what is stated in the null hypothesis
 - **What we want to prove lies with H_a**

OUR GOAL IS TO DISPROVE H_0

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Making the “Belief” Decision

- Use Mary’s report to test the claim of the firm’s manager:
 H_0 : The average amount due to be paid per account was at least \$300.

- You want to choose the phrase which best fills the blanks

The data, *all by itself*, makes me _____ suspicious, because the data, *all by itself*, _____ contradicts the statement above!”

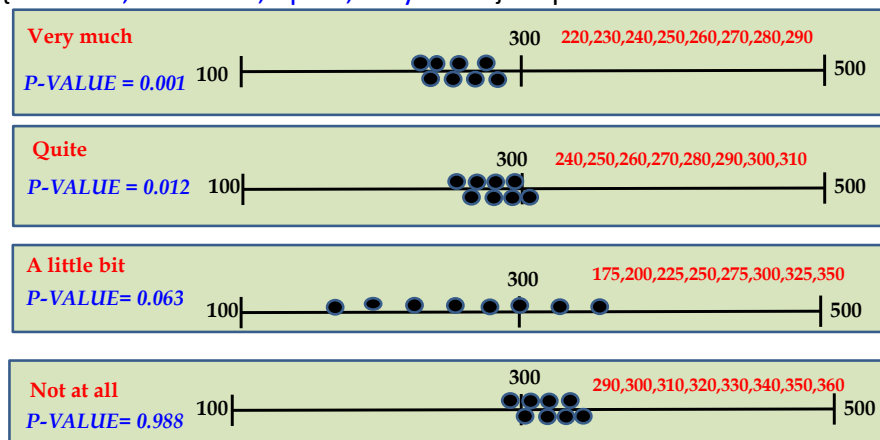
{not at all, a little bit, quite, very much}

OUR GOAL IS TO DISPROVE H_0

Evaluating evidence against H_0

The average amount due to be paid per account was at least \$300.

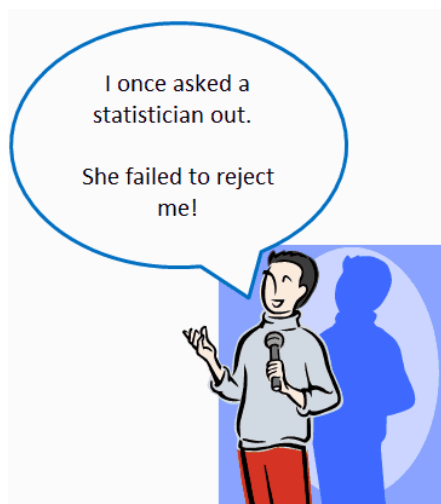
{not at all, a little bit, quite, very much} suspicious



p-value: measure of significance

P-value is the probability of observing data as extreme as, or more extreme than, the data actually observed, under the assumption that the statement is true.

P-value range	How suspicious are you about the claim?
P-value ≤ 0.01	Very much suspicious: Very strong evidence against the statement
$0.01 < \text{P-value} \leq 0.05$	Quite suspicious Moderately strong evidence against the statement
$0.05 < \text{P-value} \leq 0.10$	A little bit suspicious Rather weak evidence against the statement
P-value > 0.10	Not at all suspicious: Very little or no evidence against the statement

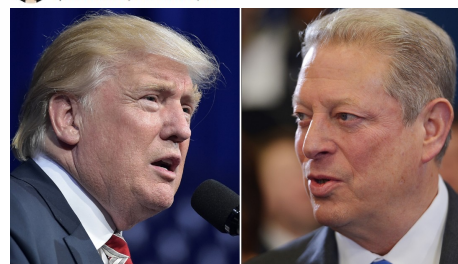


Steps in hypothesis testing

- Set-up Null hypothesis; Alternative hypothesis
 - **Choose level of significance α**
(usually 1% or 5%)
 - **Compute p-value**
 - **Decision**
 - If $p\text{-value} \leq \alpha$ reject Null hypothesis
 - If $p\text{-value} > \alpha$ retain Null hypothesis/ **fail to reject Null**
- do NOT say accept Null Hypothesis!!!**

Donald Trump meets with Al Gore on climate change

By Tal Kopan, CNN
Updated 4:51 PM ET, Mon December 5, 2016



Is Global Warming Fake News?

H_0 : Global warming is fake news
 H_a : Global Warming is real

$\alpha = 5\%$
Reject H_0 when $p \leq 0.05$

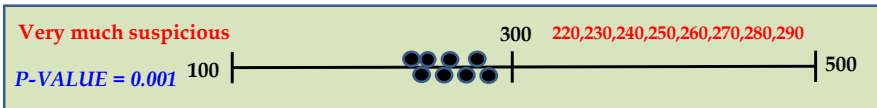
$\alpha = 1\%$
Reject H_0 when $p \leq 0.01$

Which politician choose which level of significance?

*Gore, an environmental activist, won 2007 Nobel Peace Prize for work on climate change.

Apply steps of hypothesis testing for this evidence ($\alpha=1\%$; $\alpha=5\%$)

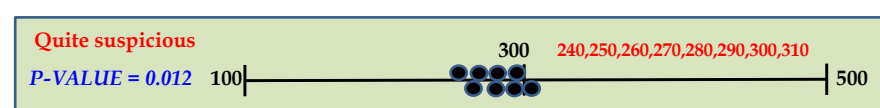
H_0 : The average amount due to be paid per account was at least \$300.



Conclusion?

Apply steps of hypothesis testing for this evidence ($\alpha=1\%$; $\alpha=5\%$)

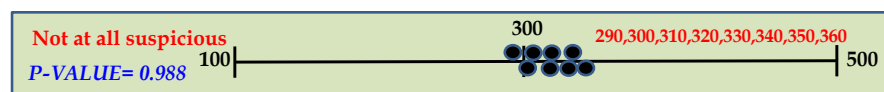
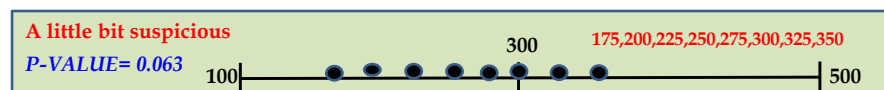
H_0 : The average amount due to be paid per account was at least \$300.



Conclusion?

Apply steps of hypothesis testing for this evidence ($\alpha=1\%$; $\alpha=5\%$)

H_0 : The average amount due to be paid per account was at least \$300.



Conclusion?

Errors in Hypothesis Testing

Since hypothesis tests are based on sample data, we must allow for the possibility of errors.

– Type I Error and Type II Error

Conclusion \ Population	Population	
	H_0 True	H_a True
Not Reject H_0	Correct Conclusion	Type II Error (β)
Reject H_0	Type I Error (α)	Correct Conclusion

– There is a trade-off between Type-I and Type-II risk: One increases (decreases), the other decreases (increases).

Level of significance and error types

- **Fix level of significant α , if we follow the decision rule; Type 1 error will be α**
- Before testing, to control Type-I risk to a small pre-determined level:
 - Choose a maximum value that we (i.e., manager) can live with; e.g., $\alpha = 1\%$, 5% , or very rarely 10% .

Example

A new teaching method is developed that is believed to be better than the current method.

Research hypothesis

- Alternative Hypothesis:
 - The new teaching method is better.
- Null Hypothesis:
 - The new method is no better than the old method.

How to Choose H_0 and H_a ?

What we want to “prove” is H_a .

Sometimes, it is not clear how to set up H_0 , H_a . Usually there are two types of Hypothesis tests.

- **Research hypothesis**
 - **H_a** : Prove that **new** research or policy works well
- **Challenge hypothesis**
 - **H_a** : Prove that an **old, long-standing** claim is wrong

A new drug is developed with the goal of lowering blood pressure more than the existing drug.

Research hypothesis

- Alternative Hypothesis:
 - The new drug lowers blood pressure more than the existing drug.
- Null Hypothesis:
 - The new drug does not lower blood pressure more than the existing drug.

The label on a soft drink bottle states that it contains 67.6 fluid ounces (2 liters), the manufacturer would like to test it.

Challenge Hypothesis

- Alternative Hypothesis:
 - The label is incorrect: $\mu \neq 67.6$ ounces.
- Null Hypothesis:
 - The label is correct: $\mu = 67.6$ ounces.



Problem 1

In the past, 75% of the tourists who visited Miami went to see Miami Beach. The management of Miami Beach recently undertook a promotional campaign. They are interested in determining whether the campaign actually **increased** the proportion of tourists visiting Miami Beach.

- What is the correct set of hypotheses?
- p-value is 0.03 what is your conclusion?
- Assume that in reality the promotional campaign actually **increased** the proportion of tourists visiting Miami Beach. Did your decision in part b make any error? If yes, what type of error? **What if p-value=0.3?**

Problem 2

Your investment executive claims that the average yearly rate of return on the stocks she recommends is at least 10.0%. You plan on taking a sample to test her claim.

- What is the correct set of hypotheses?
- p-value is 0.03 what is your conclusion?
- Assume that in reality, the average yearly rate of return on the stocks she recommends is indeed at least 10.0%. Did your decision in part b make any error? If yes, what type of error? **What if p-value=0.3?**

Part II: Math Details of Hypothesis Testing

- How to compute P-values given sample data?
- Steps in hypothesis testing

Example 1: Loan Application

Null Hypothesis: The average amount due to be paid per account was at least \$300

$$H_0 : \mu \geq 300$$

$$H_A : \mu < 300$$

Data: sample mean of 30 accounts is 283

The data by itself makes me *{not at all, a little bit, quite, very much}* suspicious

- P-value = ?

p-value: measure of significance

P-value is the probability of observing data as extreme as, or more extreme than, the data actually observed, under the assumption that the statement is true.

P-value range	How suspicious are you about the claim?
P-value ≤ 0.01	Very much suspicious: Very strong evidence against the statement
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$0.05 < \text{P-value} \leq 0.10$	A little bit suspicious Rather weak evidence against the statement
P-value > 0.10	Not at all suspicious: Very little or no evidence against the statement

$$P(\bar{X} \leq 283 | H_0) = P(\bar{X} \leq 283 | \mu \geq 300)$$

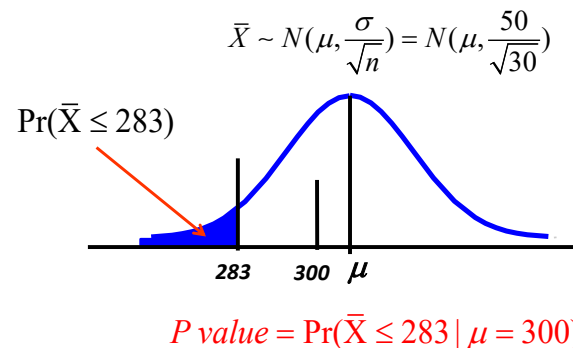
How to compute p-value?

- Null Hypothesis:
 - The average amount due to be paid per account was at least \$300
- Data:
 - sample mean of 30 accounts is 283
 - population standard deviation σ is known = 50
- P-value:

$$P(\bar{X} \leq 283 | H_0) = P(\bar{X} \leq 283 | \mu \geq 300)$$

Compute p-value Assume σ is known = 50

- Null hypothesis: $\mu \geq 300$
- Sampling distribution



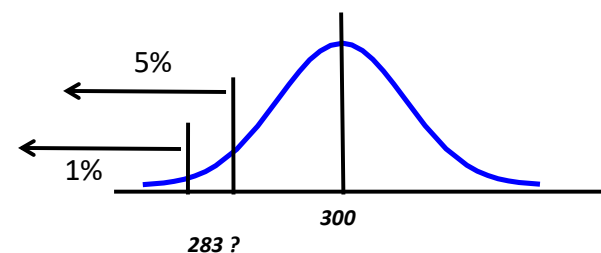
$$P \text{ value} = \Pr(\bar{X} \leq 283)$$

$$\bar{X} \sim N(300, \frac{50}{\sqrt{30}})$$

$$z\text{-score of } 283 =$$

$$p\text{-value} =$$

Alternative way: critical values



Example 2: Metro EMS

- A major west coast city provides one of the most comprehensive emergency medical services in the world. Operating in a multiple hospital system with approximately 20 mobile medical units, the city has maintained the service goal to respond to medical emergencies with a mean time of **12 minutes or less**.
- The manager of this service wanted to test if the 12 minutes standard still hold. A sample of **40** response times was collected, and the average response time was **13.25** minutes. **Population standard deviation is assumed to be 3.2 minutes.**
- What can be concluded regarding the average response time of the city's medical emergencies?
 - with 0.05 level of confidence?
 - with 0.01 level of confidence?

Example 2: Metro EMS

- Set-up Null hypothesis & Alternative hypothesis



Where μ : mean response time for the population of medical emergency requests.

- Choose level of significance $\alpha=0.05$

Example 2: Metro EMS

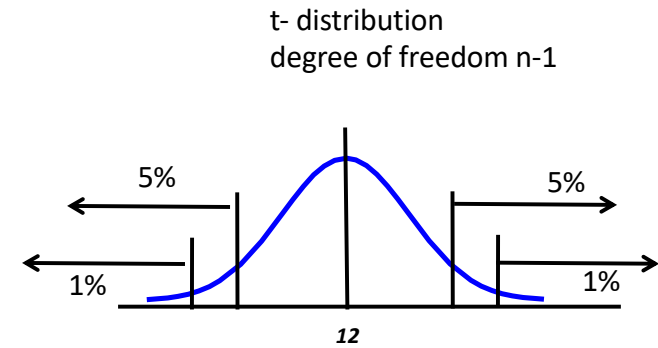
- P-value

Decision using critical values

Example 2: Metro EMS

- A major west coast city provides one of the most comprehensive emergency medical services in the world. Operating in a multi-hospital system with approximately 20 mobile medical units, the city has maintained the service goal to respond to medical emergencies with a mean time of **12 minutes or less**.
- The manager of this service wanted to test if the 12 minutes standard still hold. A sample of **40** response times was collected and the average response time was **13.25 minutes**. ~~Population standard deviation is assumed to be 3.2 minutes.~~ Sample data
- Sample mean = 12.535. sample standard deviation = 4.18.
- What can be concluded regarding the average response time city's medical emergencies?
 - with 0.05 level of confidence?
 - with 0.01 level of confidence?

15.3
20.8
11.3
9.3
14.3
19.5
12.3
13.1
15.8
9.5
8.2
6
9.6
10.5
12
23.7
11.4
5.5
9.4
14.9
13.1
13
9.8
10.5
18.6
11.2
11.1
20.1
17.7
10.5
5.5
6.7
14.4
12.8
11.7
12
13.2
16.6
9.6
10.9



Compute p-value where **σ is not known**

- Sampling distribution

$$\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t\text{-distribution}; \text{dof} = n - 1$$

- P-value=????

– Using Excel **t.dist(x, degree of freedom, 1)**

Summary for this example

- Set-up Null hypothesis; Alternative hypothesis
- Choose level of significance α (1% or 5%)
- Compute p-value
- Decision
 - If $p\text{-value} \leq \alpha$ reject Null hypothesis
 - If $p\text{-value} > \alpha$ fail to reject Null hypothesis

Example 3: Two sided hypothesis Glow Toothpaste

- The production line for Glow toothpaste is designed to fill tubes of toothpaste with a target mean weight of 6 ounces.
- Periodically, a sample of 30 tubes will be selected in order to check the filling process. Quality assurance procedures call for the **continuation of the filling process** if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; **otherwise the filling process will be stopped and adjusted**.
- One day, the sample mean is found to be 6.1 ounces; **sample standard deviation is .2 ounces**. The floor manager does not want to stop the process and argue that the .1 ounce difference between the sample mean and target is **"not a big deal."** What do you think?

Type of Hypothesis for Testing One Population Mean

- Let μ be the unknown population mean.
- Hypotheses (where μ_0 is a hypothesized value for μ):

$H_0: \mu = \mu_0$ vs. $H_a: \mu \neq \mu_0$ (two-tailed).

$H_0: \mu \geq \mu_0$ vs. $H_a: \mu < \mu_0$ (left-tailed/ lower-tailed).

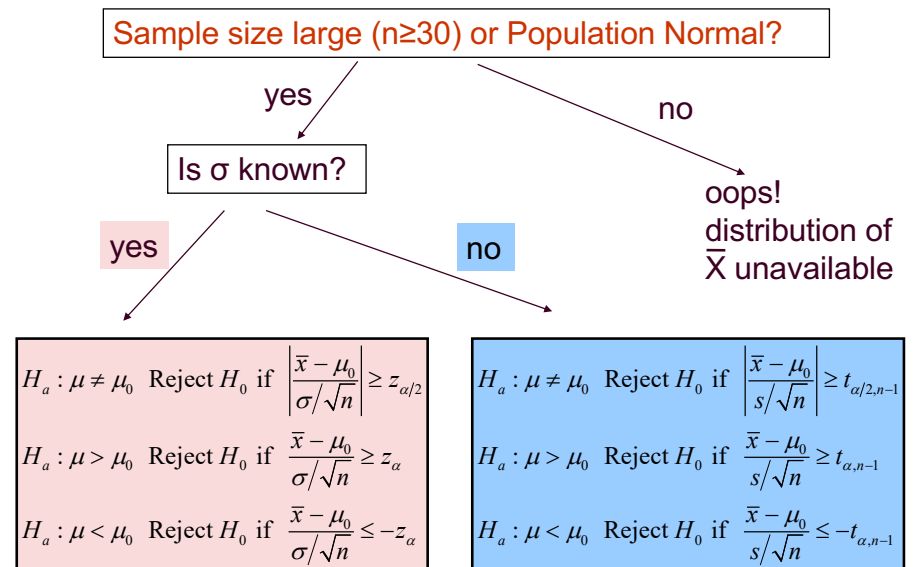
$H_0: \mu \leq \mu_0$ vs. $H_a: \mu > \mu_0$ (right-tailed/ upper-tailed).

- The equality always appears in the null hypothesis.

Steps in hypothesis testing

- Set-up Null hypothesis & Alternative hypothesis
- Choose level of significance α
- Use critical value
- Decision

Sampling Distributions and Statistical Inference for Population Mean



Level of Significance

- Defines unlikely values of sample statistic if null hypothesis is true
- Denoted by α
- Typical values are 1%, 5%, 10%
- Commonly used z values

NORMSINV(1-0.025)

α	z_α	$z_{\alpha/2}$
1%	2.326	2.576
5%	1.645	1.960
10%	1.282	1.645

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Null and Alternative Hypotheses About a Population Proportion

- The equality part of the hypotheses always appears in the null hypothesis.
- Hypothesis about a population proportion p (where p_0 is the hypothesized value of the population proportion).

$$H_0: p \geq p_0$$

$$H_0: p \leq p_0$$

$$H_0: p = p_0$$

$$H_a: p < p_0$$

$$H_a: p > p_0$$

$$H_a: p \neq p_0$$

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Tests About a Population Proportion: Large-Sample Case ($np \geq 5$ and $n(1-p) \geq 5$)

- Test Statistic:
$$z = \frac{\bar{p} - p_0}{\sigma_{\bar{p}}}$$

where

$$\sigma_{\bar{p}} = \sqrt{\frac{p_0(1-p_0)}{n}}$$

- P-value:.....

- Rejection Rule:

One-Tailed

Two-Tailed

$$H_a: p > p_0$$

Reject H_0 if $z \geq z_\alpha$

$$H_a: p < p_0$$

Reject H_0 if $z \leq -z_\alpha$

$$H_a: p \neq p_0$$

Reject H_0 if $|z| \geq z_{\alpha/2}$

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Example 4: Transaction Errors

You're an accounting manager. A year-end audit showed 6% of transactions had errors. You implement new procedures. A random sample of 500 transactions had 25 errors. Has the proportion of incorrect transactions decreased at the .05 level?

$$H_0: p \geq 0.06$$

$$H_a: p < 0.06$$

$$z = \frac{\bar{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.05 - 0.06}{\sqrt{\frac{0.06(1-0.06)}{500}}} = -0.94$$

$$z_{.05} = 1.645$$

$$-1.645 < -0.94$$

There is not sufficient evidence to conclude that the proportion of incorrect transactions decreased at the .05 significant level.

EXAMPLES, PRACTICE QUESTIONS

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Summary: Steps in hypothesis testing

- Set-up Null hypothesis; Alternative hypothesis
- **Choose level of significance α (1% or 5%)**
- Compute p-value
- Decision
 - If $p\text{-value} \leq \alpha$ reject Null hypothesis
 - If $p\text{-value} > \alpha$ fail to reject Null hypothesis
- Or making decision by critical values